The Full BKM Tail Inequality on a Top-Walsh Fiber

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Setup. For each coordinate $i \in [n]$, let $P_{i,0}, P_{i,1}$ be rank-one projectors on a real qubit and put

$$s_i = \sqrt{1 - |\langle \psi_{i,0}, \psi_{i,1} \rangle|^2}, \qquad z = \prod_{i=1}^{n} s_i.$$

For $x \in \{0,1\}^n$, write $P_x = \bigotimes_i P_{i,x_i}$, and extend the pure-state channel linearly to signed vectors by

$$\Phi(a) = \sum_x a_x P_x.$$

Let q be a strictly positive probability law, let

$$e_x = (-1)^{x_1 + \cdots + x_n}, \qquad d = ke, \qquad r = q + td > 0,$$

and set $C = \Phi(r)$, $Q = \Phi(q)$, and $H = \Phi(d)$. For $L \geq 0$, define the quantum BKM tail

$$\mathsf{Q}_L = \int_L^\infty \operatorname{Tr}\big[H(C + \lambda I)^{-1} H (C + \lambda I)^{-1}\big]\, d\lambda. \tag{1}$$

The corresponding matched-BEC tail is

$$\mathsf{E}_L = \mathbb{E}_R \sum_y \frac{d_R(y)^2}{r_R(y) + L q_R(y)}, \tag{2}$$

where R reveals coordinate i independently with probability s_i, and a_R denotes the marginal of a signed vector a.

Theorem 1 (Top-Walsh BKM tail)**.** *Under the preceding assumptions,*

$$\mathsf{Q}_L \leq \mathsf{E}_L \qquad \text{for every } L \geq 0. \tag{3}$$

In fact, all proper marginals of d vanish and hence

$$\mathsf{E}_L = z k^2 \sum_x \frac{1}{r_x + L q_x}. \tag{4}$$

Proof. We first assume $0 < s_i \leq 1$ for every i; zero parameters follow by continuity (and then both sides vanish).

For two pure qubit states, $P_{i,0} - P_{i,1}$ has eigenvalues $\pm s_i$. Thus

$$J_i := \frac{P_{i,0} - P_{i,1}}{s_i}$$

is a Hermitian unitary on their two-dimensional span. Consequently, with $J = \bigotimes_i J_i$,

$$H = k \bigotimes_i (P_{i,0} - P_{i,1}) = kzJ. \tag{5}$$

Put $A = C + LI$. Changing variables $\lambda = L + \mu$ in (1) shows that Q_L is the BKM quadratic form at A:

$$\mathsf{Q}_L = \operatorname{Tr}\big[H\mathcal{T}_A(H)\big], \qquad \mathcal{T}_A(X) = \int_0^\infty (A + \mu I)^{-1} X (A + \mu I)^{-1} d\mu.$$

In an eigenbasis of A, its kernel is

$$k_{\mathrm{BKM}}(a, b) = \frac{\log a - \log b}{a - b} \le \frac{1}{\sqrt{ab}}, \tag{6}$$

where (6) is the scalar inequality that the logarithmic mean is at least the geometric mean. Therefore

$$\mathsf{Q}_L \le \operatorname{Tr}(A^{-1/2} H A^{-1/2} H) = k^2 z^2 \operatorname{Tr}(X_A J), \qquad X_A = A^{-1/2} J A^{-1/2}. \tag{7}$$

Since J is unitary, trace-norm duality gives

$$\operatorname{Tr}(X_A J) \le |\operatorname{Tr}(X_A J)| \le \|X_A\|_1. \tag{8}$$

The matrix $Q = \Phi(q)$ is a density operator, so $0 \le Q \le I$. It follows that

$$A = C + LI \succeq C + LQ =: B. \tag{9}$$

Let $K = A^{-1/2} B^{1/2}$. Equation (9) implies $KK^* = A^{-1/2} B A^{-1/2} \preceq I$, so $\|K\|_\infty \le 1$. Moreover,

$$X_A = K X_B K^*, \qquad X_B = B^{-1/2} J B^{-1/2}.$$

Hence the trace norm is contractive:

$$\|X_A\|_1 \le \|X_B\|_1. \tag{10}$$

It remains to evaluate the latter trace norm. Let Ψ be the square signal-synthesis matrix whose xth column is $\bigotimes_i |\psi_{i,x_i}\rangle$. The local signal pairs are linearly independent, so Ψ is invertible. We have

$$B = \Psi D_{r+Lq} \Psi^*, \qquad J = z^{-1} \Psi D_e \Psi^*. \tag{11}$$

The Hermitian matrix X_B is similar to $B^{-1}J$. Congruence by the invertible synthesis matrix in (11) shows that its eigenvalues are the generalized eigenvalues of the pair (J, B), namely

$$\operatorname{spec}(X_B) = \left\{ \frac{e_x}{z(r_x + Lq_x)} : x \in \{0,1\}^n \right\}. \tag{12}$$

Thus

$$\|X_B\|_1 = \frac{1}{z} \sum_x \frac{1}{r_x + Lq_x}. \tag{13}$$

Combining (7), (8), (10), and (13) yields

$$\mathsf{Q}_L \le zk^2 \sum_x \frac{1}{r_x + Lq_x}. \tag{14}$$

Finally, summing the parity character over any missing coordinate gives $d_R = 0$ for every proper $R \subsetneq [n]$. Only the full reveal contributes to (2), with probability z, and $d_x^2 = k^2$. This proves (4), so (14) is exactly the desired inequality. □

Lemma 2 (A universal trace-norm envelope)**.** *Let q, r be arbitrary strictly positive probability laws, let d be an arbitrary real signed vector, and put*

$$C = \Phi(r), \qquad Q = \Phi(q), \qquad H = \Phi(d).$$

Then the quantum tail in (1) satisfies

$$\mathsf{Q}_L \le \sum_{j=1}^{2^n} h_j \alpha_j \le \|H\|_\infty \sum_x \frac{|d_x|}{r_x + Lq_x}. \tag{15}$$

Here $h_1 \ge \cdots \ge h_{2^n}$ are the singular values of H, and $\alpha_1 \ge \cdots \ge \alpha_{2^n}$ is the decreasing rearrangement of $\{|d_x|/(r_x + Lq_x)\}_x$.

Proof. The geometric-kernel bound (6), now with $X_A = A^{-1/2} H A^{-1/2}$, gives

$$\mathsf{Q}_L \le \operatorname{Tr}(X_A H).$$

As in (9)–(10), $A = C + LI \succeq B = C + LQ = \Phi(r + Lq)$ and $X_A = K X_B K^*$ for a contraction K, where $X_B = B^{-1/2} H B^{-1/2}$. Hence every Ky Fan norm contracts, or equivalently $s(X_A) \prec_w s(X_B)$. The generalized eigenvalues of (H, B) are $d_x/(r_x + Lq_x)$, so the singular values of X_B are the α_j's. Von Neumann's trace inequality followed by Abel summation gives

$$\operatorname{Tr}(X_A H) \le \sum_j h_j s_j(X_A) \le \sum_j h_j \alpha_j.$$

Finally $h_j \le \|H\|_\infty$, which proves the second inequality in (15). □

Lemma 3 (The zero-cutoff comparison implies every cutoff)**.** *For fixed Hermitian H, the BKM quadratic form*

$$g_A(H) = \int_0^\infty \operatorname{Tr}\big[H(A + uI)^{-1} H (A + uI)^{-1}\big]\, du$$

is decreasing in the Loewner order of the positive base A. Consequently, if the $L = 0$ BKM–BEC metric comparison is known at every (possibly subnormalized) positive input base, then its full tail comparison holds for every $L \ge 0$.

Proof. Along $A_t = A + tK$ with $K \succeq 0$, put $R = (A_t + uI)^{-1}$. Differentiation under the integral and cyclicity of trace give

$$\frac{d}{dt} g_{A_t}(H) = -2 \int_0^\infty \operatorname{Tr}(KRHRHR)\, du \le 0,$$

because $RHRHR = R^{1/2}(R^{1/2} H R^{1/2})^2 R^{1/2} \succeq 0$. For the tail, take $A = C + LI$ and $B = C + LQ = \Phi(r + Lq)$. Since $Q \preceq I$, $A \succeq B$, and hence

$$\mathsf{Q}_L = g_A(H) \le g_B(H).$$

Applying the zero-cutoff comparison at the base $r + Lq$ gives exactly

$$g_B(H) \le \mathbb{E}_R \sum_y \frac{d_R(y)^2}{r_R(y) + Lq_R(y)} = \mathsf{E}_L.$$

Normalization is immaterial because both quadratic forms have the same degree -1 homogeneity in the base. □

Corollary 4 (Relative contraction on a top-interaction fiber)**.** *If $p-q = ke$ and the segment $q+u(p-q)$ is positive, then*

$$D(\Phi(p)\|\Phi(q)) \le zD(p\|q). \tag{16}$$

In particular, for two bits this applies to every fixed-marginal fiber.

Proof. Along $r_u = q + u(p-q)$, the second derivative of the quantum relative entropy is the $L = 0$ BKM form in Theorem 1, while the second derivative of $zD(r_u\|q)$ is $zk^2 \sum_x 1/r_u(x)$. Theorem 1 compares these second derivatives. Both relative entropies and their first derivatives vanish at $u = 0$, so integrating twice gives (16). □

Corollary 5 (A single biased product-Walsh mode)**.** *Let $T \subseteq [n]$, let $q = q_T \otimes q_{T^c}$, and suppose*

$$p = p_T \otimes q_{T^c}, \qquad p_T - q_T = k\chi_T.$$

Then the full BKM tail comparison holds for the pair (p, q) and

$$D(\Phi(p)\|\Phi(q)) \le \left(\prod_{i\in T} s_i\right) D(p_T\|q_T).$$

In particular, this covers every single Fourier-mode ray based at an arbitrarily biased product law.

Proof. Both inputs have the same independent T^c factor. Relative entropy and the $L = 0$ BKM form tensor-cancel that factor; explicitly, $g_{A\otimes\sigma}(H \otimes \sigma) = g_A(H)$ for a density matrix σ. Theorem 1 applied on the subsystem T proves the zero-cutoff comparison, and the preceding lemma supplies every cutoff. On the BEC side all reveals of T^c sum to one and contribute zero relative entropy, leaving exactly the displayed factor $\prod_{i\in T} s_i$. □

Remark. The proof gives more than the BKM comparison at $L = 0$: the larger geometric kernel $1/\sqrt{ab}$ is already bounded by the matched classical Fisher information on a top-Walsh tangent. The key extra feature is that the quantum tangent is a scalar multiple of a Hermitian unitary.

Where the extension to several interactions breaks. Lemma 2 isolates the special feature used by the theorem. For a conditional interaction

$$d(x) = \chi_T(x_T)g(x_{T^c}),$$

one has the exact factorization

$$\Phi(d) = \left(\prod_{i\in T} s_i\right) J_T \otimes \Phi_{T^c}(g). \tag{17}$$

The first factor is unitary, but the second generally has non-flat singular values. Thus the two factors in the trace-norm envelope (15) no longer collapse to the same weighted ℓ_2 expression. If a general fixed-marginal tangent is decomposed into such interaction fibers, applying the triangle inequality to either $\|\Phi(d)\|_\infty$ or the trace norm of the generalized score introduces cross terms of size $\sqrt{E_i E_j}$; summing only the diagonal fiber bounds can therefore lose a factor equal to the number of active fibers. On the BEC side the missing information is present in all reveal sets $R \supseteq T$, through the quadratic forms

$$\sum_y \frac{d_R(y)^2}{r_R(y) + Lq_R(y)}.$$

Consequently, an extension must use these subset terms to control the singular-value spread in $\Phi_{T^c}(g)$ and the cross terms between different T's; a bare trace-norm triangle inequality cannot do so. At a product reference the interaction modes are orthogonal and these cross terms disappear, which is precisely why the local product-reference proof tensorizes.